

n ≤ 8 Review Summary

One-page external-review summary for the repo-local elementary n ≤ 8 theorem and selected-witness corroboration.

Status	Repo-local n ≤ 8 finite-case release packet
Global status	Falsifiable/open; no general proof or counterexample claimed
Date	2026-07-09

Purpose

This packet is ready for an external reviewer to inspect the repo-local elementary n ≤ 8 theorem and selected-witness corroboration. The global problem remains open; the request is only to review this small-case proof and its finite-case artifact.

What To Review

1. The elementary route: base-apex counting, octagon equality saturation, equilateral turns, and the C8 vertex-cover contradiction.
2. The selected-witness incidence route: n ≤ 7 by counting/crossing, n=8 by 15 canonical survivor classes and exact obstruction.
3. The focused delicate certificates: classes 3, 4, 5, and 14, with class 14 the main Groebner plus strict-convexity branch audit.
4. The independent replays: SymPy-free partial recheck and z3 NRA all-class cross-check.

Minimal Commands

```
python scripts/independent_check_n8_artifacts.py --check --json
python scripts/enumerate_n8_incidence.py --summary
python scripts/analyze_n8_exact_survivors.py --check --json
python scripts/check_n8_class14_certificate.py --check --json
```

Safe Review Outcomes

Accepted: the elementary n ≤ 8 theorem and/or selected-witness corroboration are suitable for paper-style citation with stated scope.

Gap found: record the exact failed lemma, checker, certificate, or convention; do not change global status without a separate source-of-truth update.